

Supersonic PINN vs validated modal CSV

Run: assets/pinn_supersonic/singlecase_pq_firstorder_M150_a01625_VALIDATED_anchor_ENVELOPED_ft

alpha: 0.1625

Mach: 1.5

p_rel: 0.012733

q_rel: 0.0125731

p_y_num_rel: 0.0332395

q_vs_dpred_rel: 0.0279482

align_scale_abs: 0.443736

pred_max_abs_p_raw: 1.0023

Zone errors:

center_|y|<=50: p_rel=0.005017, q_rel=0.008111, max|p_pred|=0.4448

center_|y|<=60: p_rel=0.01307, q_rel=0.00972, max|p_pred|=0.4448

diagnostic_|y|<=80: p_rel=0.03181, q_rel=0.02763, max|p_pred|=0.4448

left_-120_-80: p_rel=0.1338, q_rel=0.3436, max|p_pred|=0.1962

right_50_120: p_rel=1.245, q_rel=4.269, max|p_pred|=0.02529

right_y>80: p_rel=1.267, q_rel=3.3, max|p_pred|=0.00585

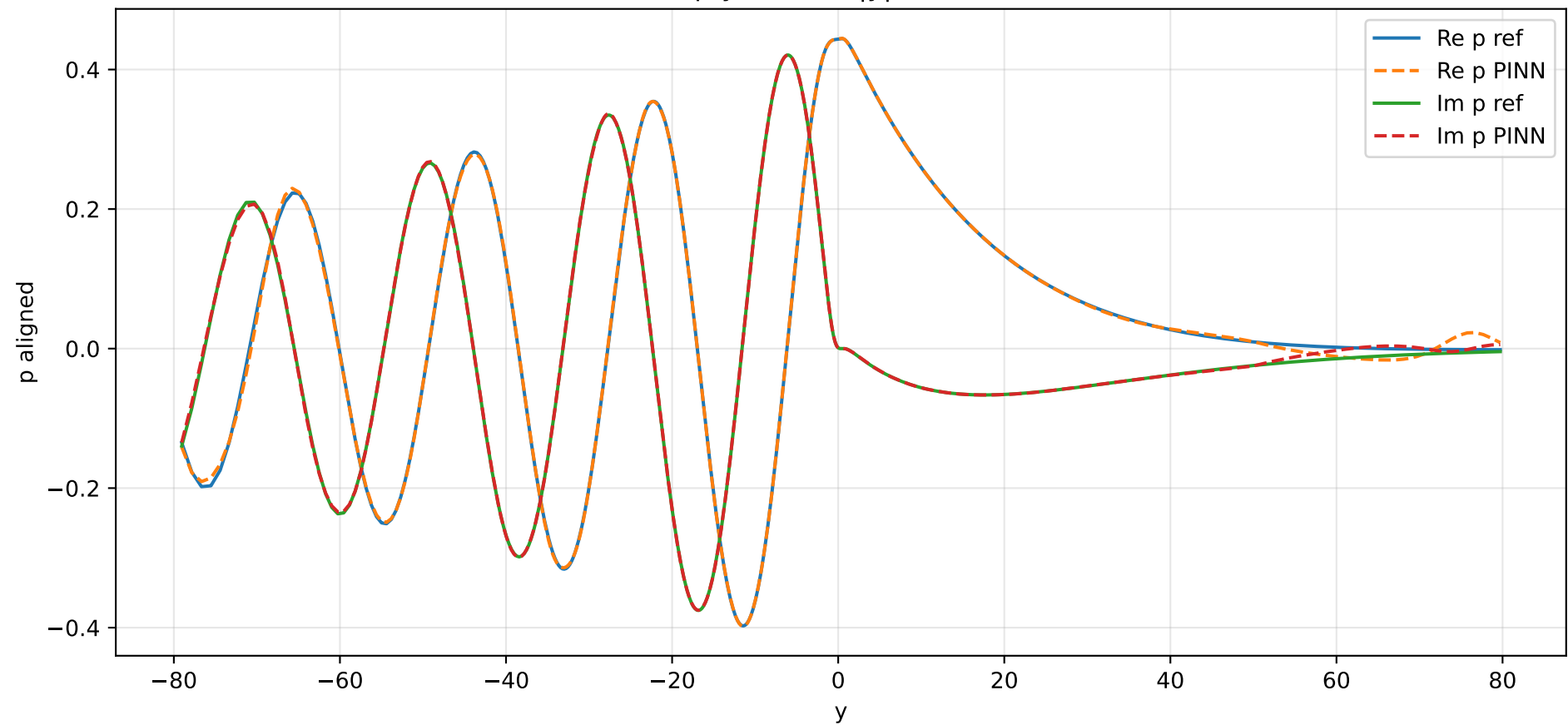
right_y>120: p_rel=0.7774, q_rel=1.141, max|p_pred|=0.0002444

left_y<-80: p_rel=0.4914, q_rel=0.4566, max|p_pred|=0.1962

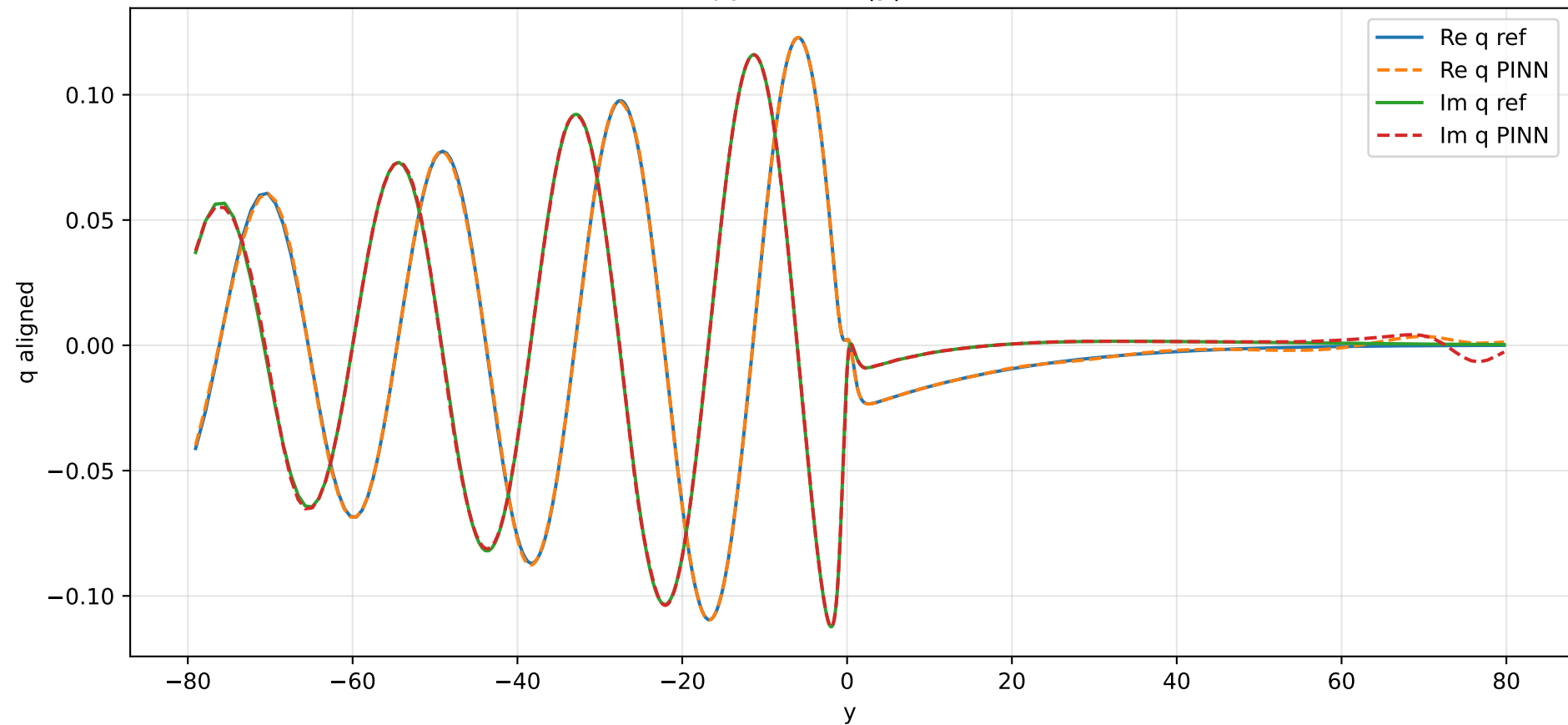
outside_|y|>80: p_rel=0.3023, q_rel=0.2537, max|p_pred|=0.1962

full_domain: p_rel=0.1937, q_rel=0.173, max|p_pred|=0.4448

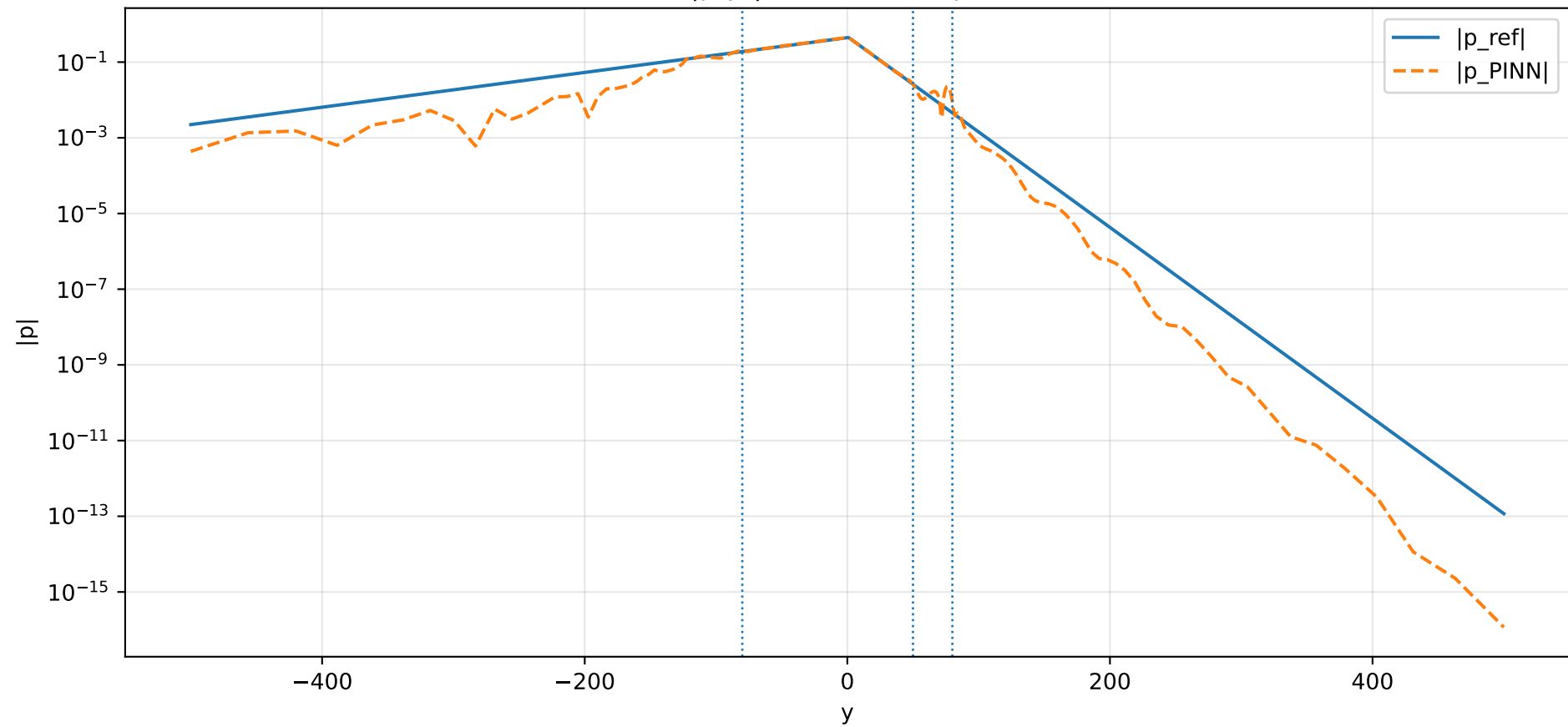
$p(y)$, centre $|y| \leq 80$



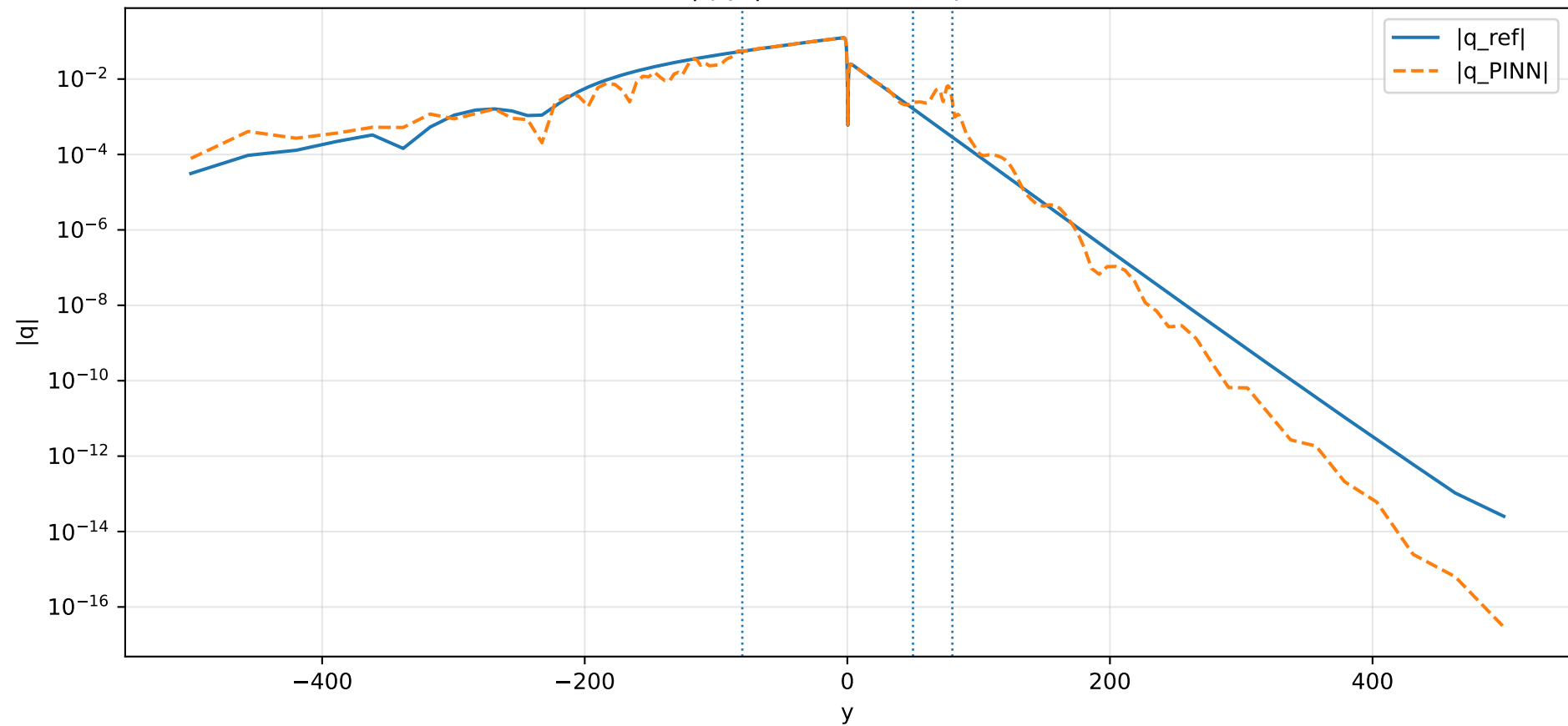
$q(y)$, centre $|y| \leq 80$

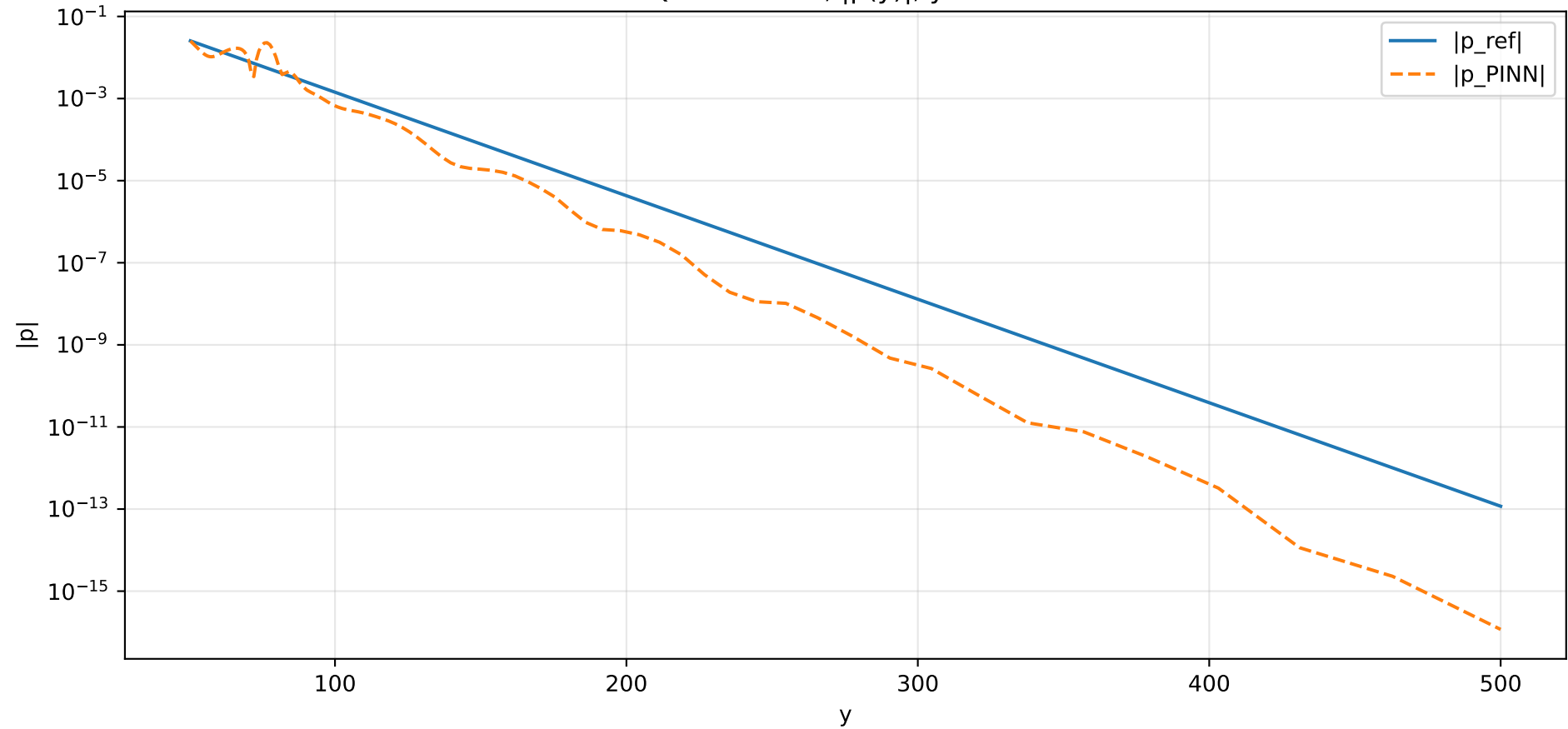


$|p(y)|$ domaine complet

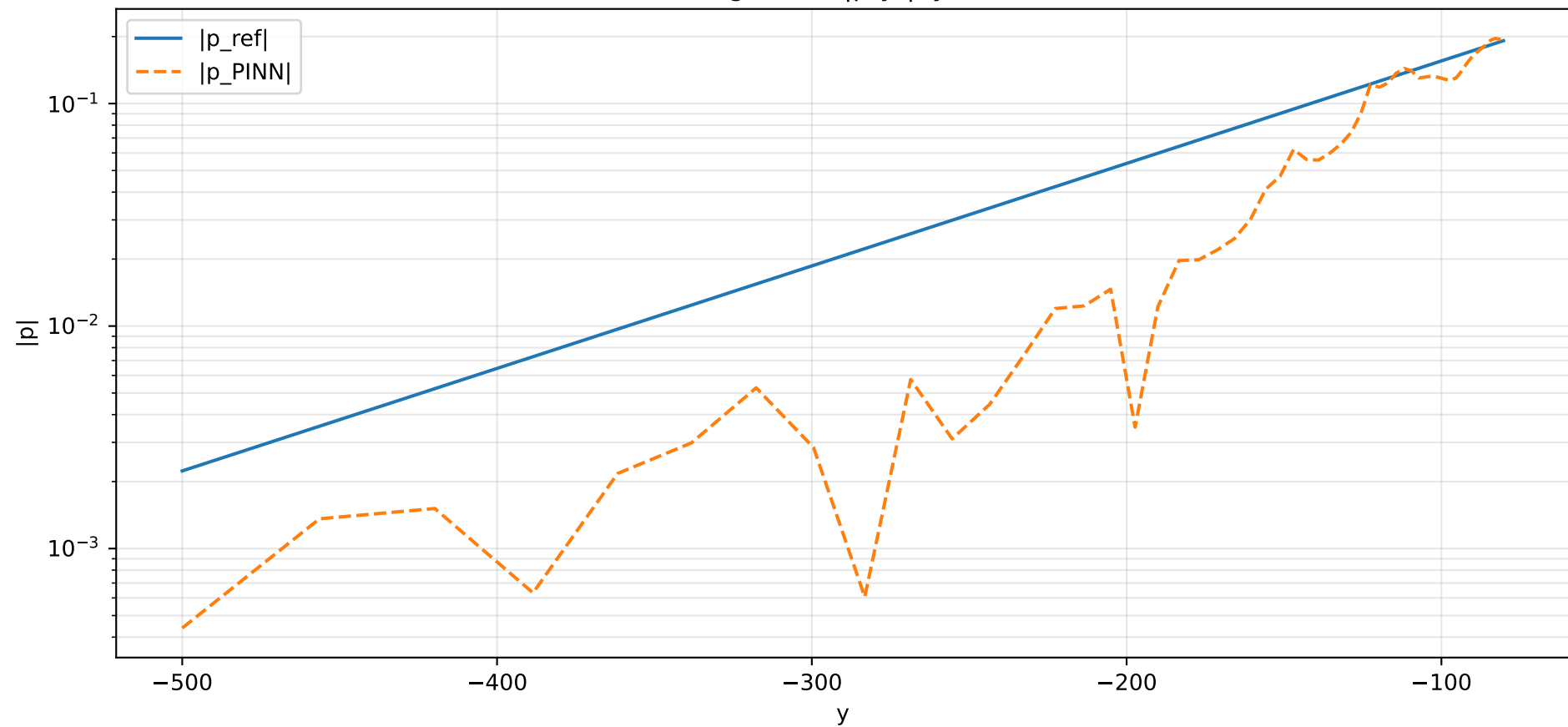


$|q(y)|$ domaine complet

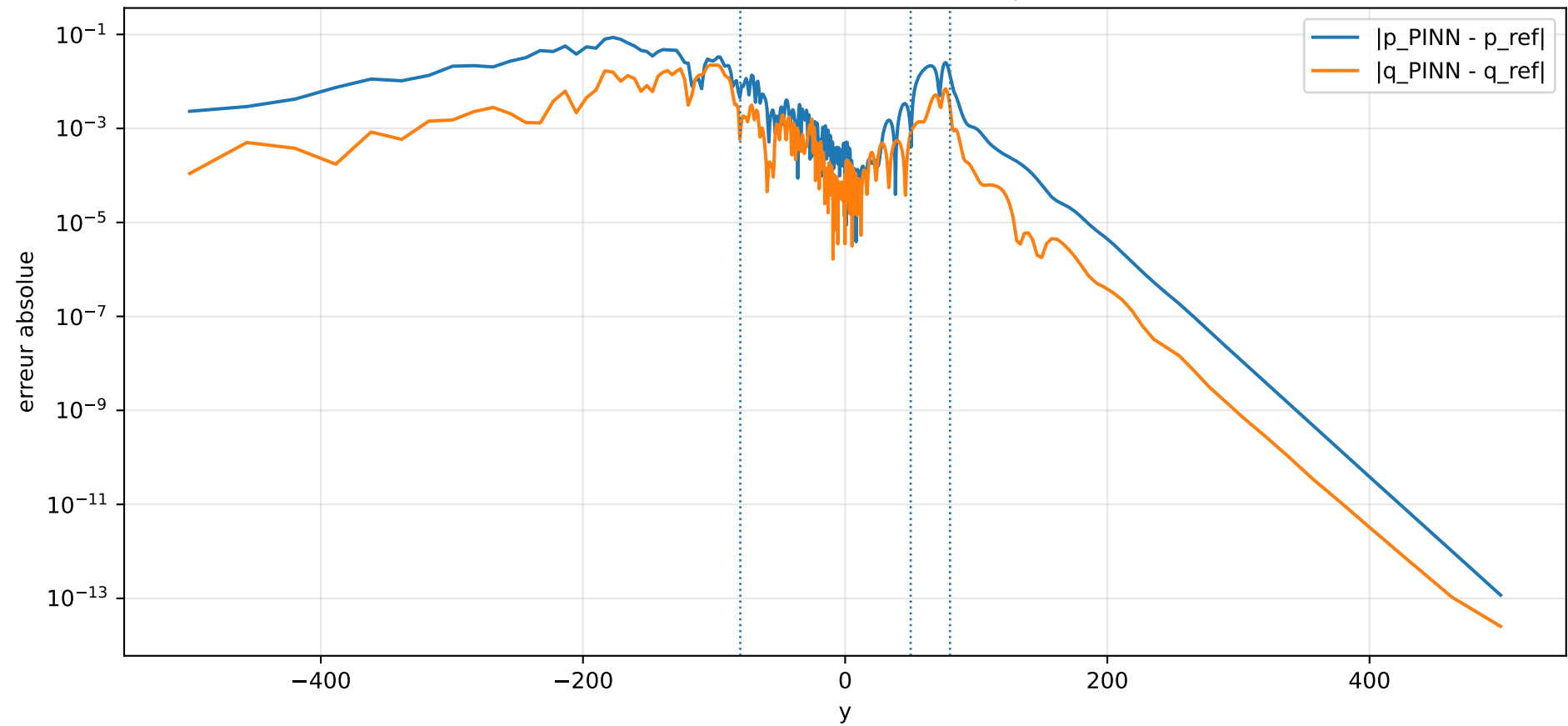


Queue droite, $|p(y)|, y \geq 50$ 

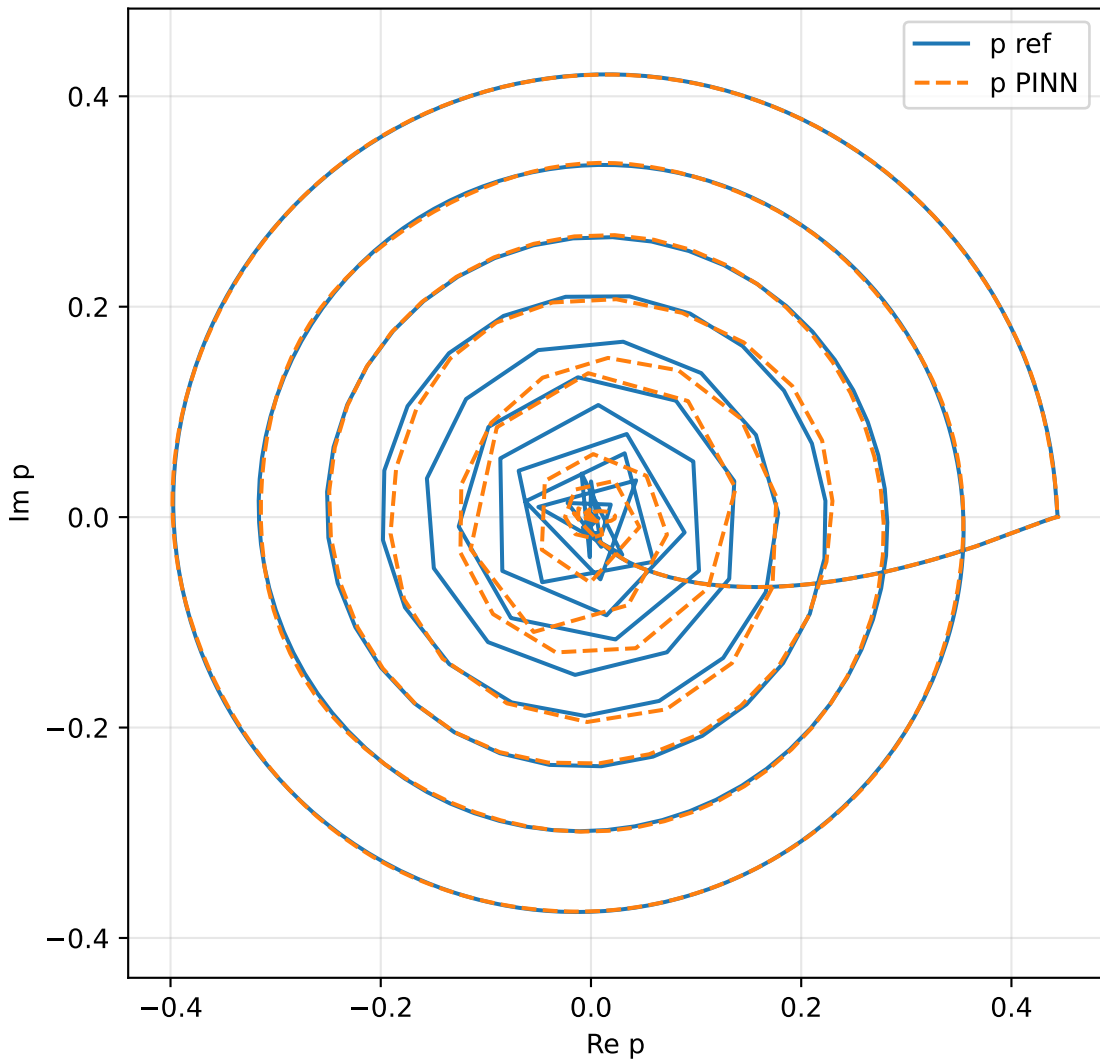
Queue gauche, $|p(y)|$, $y \leq -80$



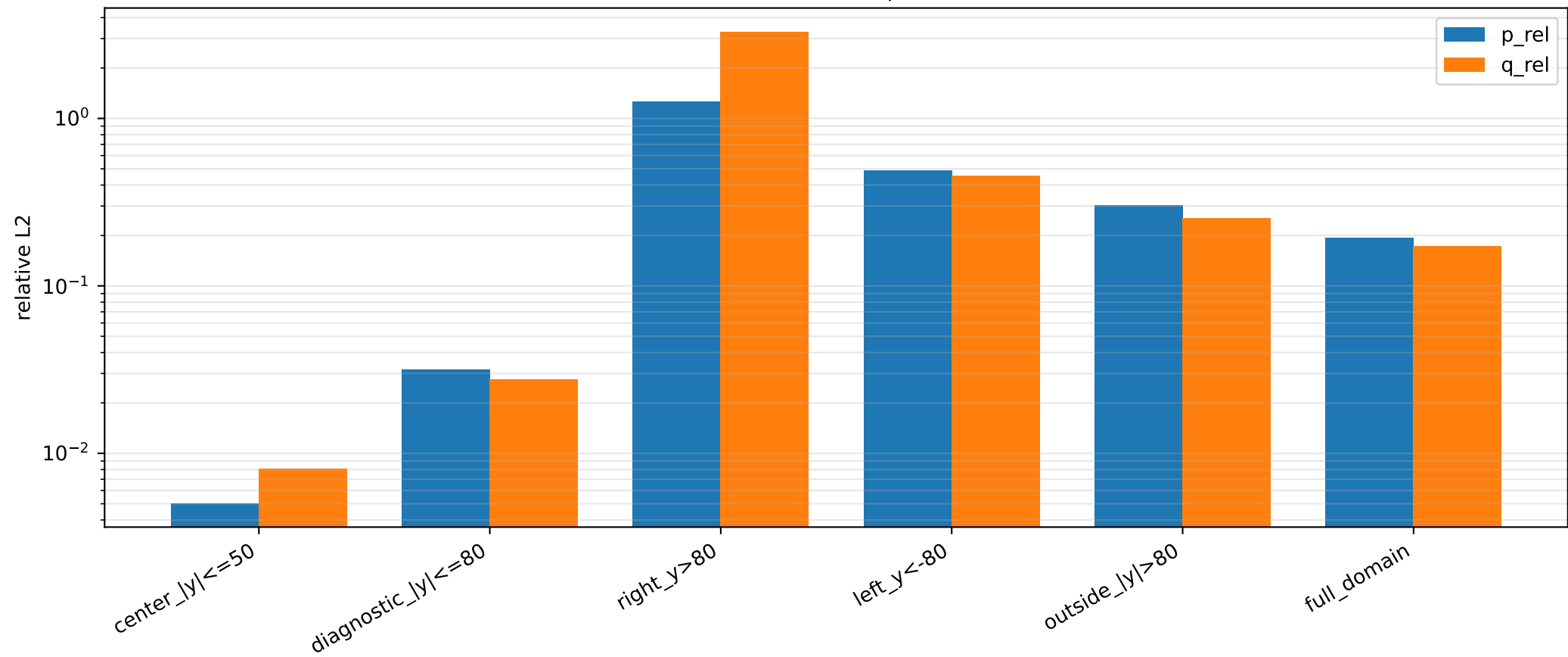
Erreur absolue domaine complet



Trajectoire complexe $p(y)$



Erreurs relatives par zone



Compatibilité $q \approx p_y$, centre $|y| \leq 80$

